

# Nonlinear Re-Randomization and CG Diversification

Why the new gadgetize reverted to the nonlinear RGs, and why each CG is now drawn from a random menu

local\_mixing project

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## Abstract

The 2026-07-20b revision of the new-gadgetize recipe makes two changes to the gadget body: the linear CNOT re-randomization gadgets  $\{RG2, RG3\}$  are replaced by the legacy *nonlinear* g57 networks  $\{RG1, RG2, RG3\}$  (one uniform draw per shared gate), and the fixed four-fragment realization of the shared computation gadget (CG) is replaced by a per-gate uniform draw over *seven* structurally different realizations. This note records the motivation: what the degree-bounded reconstruction experiments showed about the linear design, what the reversal buys and what it deliberately gives up, and why implementation diversity of the CG is itself a security measure. Companion changes (the final commuting-order rerandomization and the sandwich column float) are summarized at the end.

## 1 Setting

The two-share gadget maps a circuit  $C$  on  $n$  wires to an equivalent circuit on  $2n$  wires. Every logical value  $v$  is carried as an XOR pair  $v = s_v \oplus p_v$ ; the construction consists of random Z bookends on the aux half, a  $W_i$  encode/decode curtain of CNOTs that installs and removes the sharing, one *computation gadget* (CG) per source gate, and a stream of *re-randomization gadgets* (RGs) that churn the sharing between CGs. The zero-slice CNOT/CCNOT preblock guards the input port.

The original new-gadgetize design chose *linear* RGs (a 3-CNOT re-pairing RG2 and a 2-CNOT mask-refresh RG3) and one fixed four-fragment CG. Both choices optimized a specific adversary model: *gate-local non-completeness* — no single physical gate ever reads both carriers of one value, so every prefix of the circuit is first-order probe-masked — plus gate-count leanness.

## 2 The adversary that changed the decision

The project's security target is indistinguishability against *all feasible distinguishers*, evaluated against concrete cheap attack families. The family that drove this revision is *degree-bounded algebraic reconstruction* (`hmap_affine`): for prefixes  $C_i$  of the original and  $G_j$  of the gadgetized circuit, fit each bit of  $C_i(x)$  as a degree- $d$  GF(2) function of *all* of  $G_j$ 's wires (train/holdout split; inconsistency scores 0.5). Low error means the original computation's progress is legible at that point of the gadget; the ideal map is flat at 0.5 except the two forced I/O corners. The

measure is invariant to affine re-encodings, which is exactly what makes it the right lens for a sharing-based gadget: *the XOR sharing itself is free for this adversary*.

Two facts came out of the measurement campaign:

1. **Degree-2 reconstruction is more persistent than degree-1 under mixing.** Mixing erodes the degree-1 (affine) diagonal quickly, but the degree-2 diagonal survives much longer. The affine conjugation twists of the mixing stage cannot address this: an affine change of variables is transparent at every degree.
2. **The gadget encodings differ sharply.** The old (legacy g57) gadget’s output is *not* recoverable even by restricted degree-2 predictors, while the linear-RG CNOT gadget is affine-transparent at its interface. Computing the ANF degree of each component isolated the cause: the sharing setup  $(M, W_i)$  is affine everywhere; the nonlinearity of the legacy gadget lives in its gate application (degree 2, once per source gate, compounding) and its re-randomizers — RG1 (degree 3), RG2 (degree 2), RG3 (degree 2). The linear replacement kept the setup and *removed exactly the nonlinear part*.

The structural point is simple: affine maps compose to affine maps. With linear RGs, the entire re-randomization stream between any two CGs is one affine transformation of the wire state, so a degree- $d$  predictor sees through an *arbitrarily long* RG stream at no cost in degree. The stream churned the sharing but hid nothing from the adversary that matters. Nonlinear RGs break this composition: every RG multiplies algebraic degree into the transition, and reconstruction across a stretch of the body must now pay degree for every RG it crosses.

### 3 The reversal: nonlinear $\{RG1, RG2, RG3\}$ , one per CG

The revision reinstates the legacy g57 networks, drawn uniformly at one RG per CG gap (`--rg-frequency`, default 1 on this path):

| RG  | effect on the sharing                                           | gates | ANF degree |
|-----|-----------------------------------------------------------------|-------|------------|
| RG1 | swap the virtual values of two pairs                            | 6 g57 | 3          |
| RG2 | re-pair two pairs crosswise                                     | 6 g57 | 2          |
| RG3 | XOR $r_1 \vee \neg r_2$ of two foreign wires into both carriers | 2 g57 | 2          |

**What is bought.** Nonlinear transitions of degree 2–3 that compound along the body; a re-randomization stream that low-degree reconstruction cannot cross for free; and, as a side effect, nonlinear material (the RG3 reads of two *random* foreign wires) distributed through the entire body rather than confined to the bookends.

**What is paid, deliberately.** RG1 and RG2 gates read both carriers of a value, so gate-local non-completeness — first-order probe masking — is given up. This is a conscious ranking of adversaries: the probing model was a secondary concern, while degree-bounded reconstruction is a demonstrated, cheap, working attack on the linear design. (RG3 keeps single-carrier reads; the masking-safe CG variants below keep the option of a partially masked body alive.)

**Measured effect** ( $n=16$ , pre-mixing, old vs. revised recipe): the maps flatten (degree-1  $\sigma$   $0.125 \rightarrow 0.106$ , degree-2  $0.161 \rightarrow 0.149$ ; the interior column-mean dip that leaked the temporal layout largely evens out), and the best-case interior degree-1 reconstruction error roughly doubles

( $\approx 0.05\text{--}0.15 \rightarrow 0.2\text{--}0.3$ ). One honest caveat: the degree-2 *snapshot* diagonal persists in both versions. This is inherent to the linear XOR sharing — at any prefix the logical value is  $s \oplus p$ , a degree-1 function of the wires — independent of how nonlinear the transitions are. Erasing snapshots is the mixing stage’s job (structural DB/split mixing is what erodes it), or, in a future design, a nonlinear share *encoding*.

## 4 CG diversification: a seven-variant menu

The CG applies  $A \oplus = B \vee \neg C$  over the shares, i.e.  $f = 1 \oplus C \oplus BC$  with  $B = b_0 \oplus b_1$ ,  $C = c_0 \oplus c_1$ . The original design emitted one fixed shape — four mixed-polarity conjunction fragments — for every source gate.

**Why fixed is bad.** A fixed CG is a syntactic beacon: a fixed period, a fixed local gate-type pattern, and a fragment census that counts the source gates exactly. Regularity is precisely what static profiling and SAT-style attacks feed on — a solver that learns one CG shape has learned them all. And the fragments themselves are cold material for the mixing stage: the frozen replacement store matches g57 windows at  $\sim 99\%$  but mpmct fragment windows at  $\sim 2.8\%$ , so a fragment-heavy body resists exactly the phase-A mixing that is supposed to dissolve it.

**The menu.** Each source g57 is now emitted as one of seven realizations, drawn uniformly per gate, with the carrier roles (which carrier of  $A$  is targeted; share/pad order of  $B$  and of  $C$ ) also randomized per draw:

| # | name          | shape                        | idea                                                                                               |
|---|---------------|------------------------------|----------------------------------------------------------------------------------------------------|
| 0 | collapse both | 4 CNOT + 1 g57               | $b_0 \oplus = b_1$ , $c_0 \oplus = c_1$ put $B$ , $C$ on single wires; one g57 fires $f$ ; restore |
| 1 | collapse $c$  | 2 CNOT + 2 g57 + $\neg$ CNOT | pair sums to $BC$ ; $a \oplus = \neg c_0$ adds $1 \oplus C$                                        |
| 2 | collapse $b$  | 2 CNOT + 1 g57 + 1 frag      | $[a, B, c_0]$ plus fragment $\neg B \wedge c_1$                                                    |
| 3 | linear tail   | $\neg$ CNOT + CNOT + 4 g57   | no collapse; quad $[a, b_i, c_j]$ sums to $BC$ ; masking-safe                                      |
| 4 | legacy GADGET | 6 g57                        | the classic nonlinear network                                                                      |
| 5 | quad + pair   | 5 g57 + 1 frag               | quad $BC$ , then $(\neg c_0 \wedge c_1) + [a, c_1, c_0] = 1 \oplus C$                              |
| 6 | 4-cube ESOP   | 4 fragments                  | the previous fixed CG; masking-safe                                                                |

Vocabulary is restricted to g57s and pure conjunctions with one or two controls of any polarity. In particular, *no bare  $X$  gates*: several variants algebraically need a “ $\oplus 1$ ”, but an X census would count the source gates, so every constant is folded into a negative-control CNOT or a complemented fragment.

**Gate-by-gate.** Each realization writes only the target carrier  $a^*$  (one of  $A$ ’s two carriers, chosen per draw);  $b_0, b_1$  and  $c_0, c_1$  are the two carriers of  $B$  and of  $C$  in a per-draw random order. Notation, all over GF(2) (so  $+$  is  $\oplus$  and  $\bar{w} = 1 \oplus w$ ):  $t += w$  is a CNOT ( $t \leftarrow t \oplus w$ );  $t += \bar{w}$  a negative-control CNOT;  $t += (x \vee \bar{y})$  a g57; and  $t += (\ell \ell')$  a two-control conjunction fragment (juxtaposition = AND, each literal  $w$  or  $\bar{w}$ ). Recall  $f = 1 \oplus C \oplus BC = B \vee \bar{C}$ . Gates execute top to bottom. The same seven circuits are drawn in standard reversible-gate notation in Appendix A.

## 0 — collapse both

(4 CNOT + 1 g57):

1.  $b_0 += b_1$  *now  $b_0 = B$*
2.  $c_0 += c_1$  *now  $c_0 = C$*
3.  $a^* += (b_0 \vee \overline{c_0})$   $= B \vee \overline{C} = f$
4.  $c_0 += c_1$  *restore  $c_0$*
5.  $b_0 += b_1$  *restore  $b_0$*

## 1 — collapse $c$

(2 CNOT-type + 2 g57):

1.  $c_0 += c_1$  *now  $c_0 = C$*
2.  $a^* += (b_0 \vee \overline{c_0})$
3.  $a^* += (b_1 \vee \overline{c_0})$  *the two g57s sum to  $BC$*
4.  $a^* += \overline{c_0}$  *adds  $\overline{C} = 1 \oplus C$*
5.  $c_0 += c_1$  *restore  $c_0$*

## 2 — collapse $b$

(2 CNOT + 1 g57 + 1 fragment):

1.  $b_0 += b_1$  *now  $b_0 = B$*
2.  $a^* += (b_0 \vee \overline{c_0})$   $= 1 \oplus c_0 \oplus Bc_0$
3.  $a^* += (\overline{b_0} c_1)$   $= c_1 \oplus Bc_1$ ; *total =  $f$*
4.  $b_0 += b_1$  *restore  $b_0$*

## 3 — linear tail

( $\overline{\text{CNOT}}$  + CNOT + 4 g57; masking-safe, no carrier collapsed):

1.  $a^* += \overline{c_0}$
2.  $a^* += c_1$  *gates 1-2 give  $\overline{c_0} \oplus c_1 = 1 \oplus C$*
3.  $a^* += (b_0 \vee \overline{c_0})$
4.  $a^* += (b_1 \vee \overline{c_0})$
5.  $a^* += (b_0 \vee \overline{c_1})$
6.  $a^* += (b_1 \vee \overline{c_1})$  *the four g57s sum to  $BC$*

## 4 — legacy GADGET

(6 g57;  $c_1$  borrowed as restored scratch):

1.  $c_1 += (b_0 \vee \overline{b_1})$  *scratch on  $c_1$*
2.  $a^* += (c_1 \vee \overline{b_1})$
3.  $a^* += (b_0 \vee \overline{c_1})$
4.  $c_1 += (b_0 \vee \overline{b_1})$  *restore  $c_1$*
5.  $a^* += (b_1 \vee \overline{c_0})$
6.  $a^* += (c_0 \vee \overline{b_0})$

## 5 — quad + pair

(5 g57 + 1 fragment; no carrier collapsed):

1.  $a^* += (b_0 \vee \overline{c_0})$
2.  $a^* += (b_1 \vee \overline{c_0})$
3.  $a^* += (b_0 \vee \overline{c_1})$
4.  $a^* += (b_1 \vee \overline{c_1})$  *the four g57s sum to  $BC$*
5.  $a^* += (\overline{c_0} c_1)$
6.  $a^* += (c_1 \vee \overline{c_0})$  *gates 5–6 sum to  $1 \oplus C$*

## 6 — 4-cube ESOP

(4 fragments; the previous fixed CG, masking-safe):

1.  $a^* += (\overline{c_1} b_0)$
2.  $a^* += (c_1 \overline{b_1})$
3.  $a^* += (c_0 b_1)$
4.  $a^* += (\overline{c_0} \overline{b_0})$  *the four cubes XOR to  $B \vee \overline{C}$*

Every non-target carrier that a variant touches (the collapses in 0–2, the scratch in 4) is written an even number of times and so is restored exactly at the wire level; verified exhaustively over all 7 variants, 8 carrier role assignments, and 64 inputs.

## What diversification buys.

- *Irregularity*: no fixed CG period, no repeated shape, varying local gate-type statistics — less structure for syntactic profiling and for solvers to latch onto. The carrier-role randomization scrambles pin patterns even between same-variant draws.
- *DB-warmth*: the body census flips from fragment-heavy to g57-dominant (at  $n=16$ , g57 / CNOT / fragment counts  $822/277/707 \rightarrow 1269/486/264$ ), with CNOTs only sprinkled — and a few CNOTs within many g57s are well tolerated by the replacement store, while the g57 mass mixes at the  $\sim 99\%$  match rate.
- *More nonlinearity in the body*: five of seven variants carry g57 material into the CG itself, compounding with the nonlinear RG stream.

Cost: the menu averages  $\sim 5.1$  gates per CG versus 4, i.e.  $\sim 12\%$  body growth at  $n=16$  — accepted.

Every variant is verified exhaustively (all 7 variants  $\times$  8 role draws  $\times$  64 carrier assignments: exact update, wire-level restoration of every non-target carrier, vocabulary compliance), and the driver’s pinned zero-slice functionality check runs on every build.

## 5 Companion changes, briefly

**Commuting-order rerandomization** (`commuting_shuffle`, 5689c8ea). The construction-time layout  $S_1 | Z | W | \text{body} | W^{-1} | Z$  was legible in the heatmaps as an S-shape — no mixing between the computation block and the bookends, with the affine  $W_i$  curtain sitting between them. The emitted order is now a fresh random order that preserves only the relative order of gate pairs that *provably* collide (`XGate::collides`: “commute unless proven otherwise”, including the opposite-polarity shared-control exemption). This is a one-time, structure-dependent reorder — consistent with the principle that local mixing belongs to the mixing stage, whose job the gadget stage only sets up.

**Sandwich column float** (9323272a). In the sliced sandwich, each middle-column CNOT is assigned a registered random direction and floats that way to its commutation extreme, dissolving the  $C|N|D$  seam — the sandwich’s most structure-revealing feature — into a band before gadgetization.

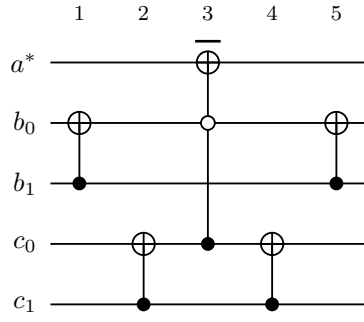
## 6 Open items

The production  $n=128$  A/B (old vs. revised recipe through the two-phase mixing pipeline, evaluated with `hmap_affine` at degrees 1 and 2) has not yet been run. The persistent degree-2 snapshot legibility of the linear XOR sharing remains the deepest open design question; the candidate answers are a nonlinear share encoding (snapshots stop being degree-1 in the wires) and generalized nonlinear RG3 injections (compensated pair-writes with arbitrary conjunction fragments).

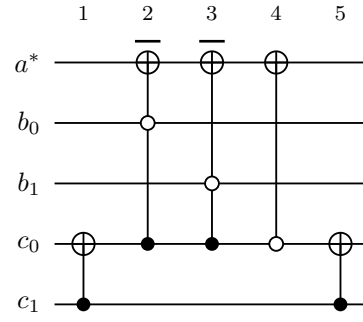
## A Gate-level diagrams of the CG menu

Each realization of Section 4 is drawn below in standard reversible-gate notation. The five horizontal lines are the carrier wires (top to bottom:  $a^*$  the targeted carrier of  $A$ , then  $b_0, b_1, c_0, c_1$ ); the untouched second carrier of  $A$  is a mask and is omitted. A gate is a vertical line joining its pins. A filled pin  $\bullet$  is a positive control (active when its wire is 1), an open pin  $\circ$  a negative control (active when 0), and  $\oplus$  the target that is flipped when the gate is active. A plain conjunction gate is active exactly when all its controls are satisfied. A **g57** gate carries a bar over its target,  $\bar{\oplus}$ : it is *complemented*, active on the complement of its control cube — realizing  $t += (x \vee \bar{y})$  from pins  $\circ$  on  $x$  and  $\bullet$  on  $y$ . Columns are numbered to match the gate sequences in Section 4 and apply left to right.

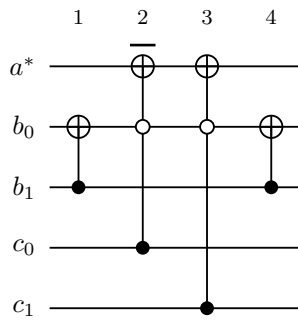
**0 — collapse both**



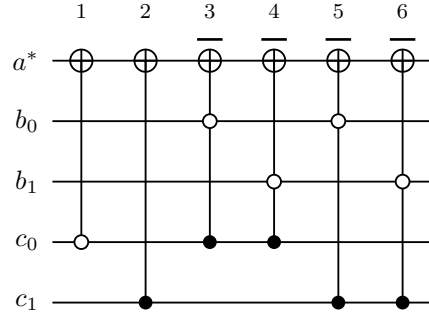
**1 — collapse  $c$**



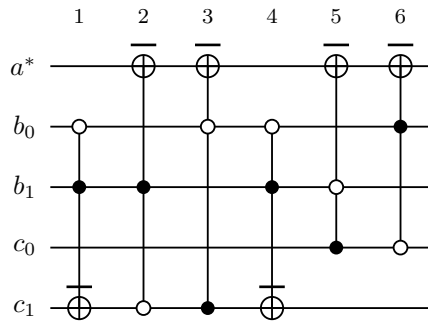
**2 — collapse  $b$**



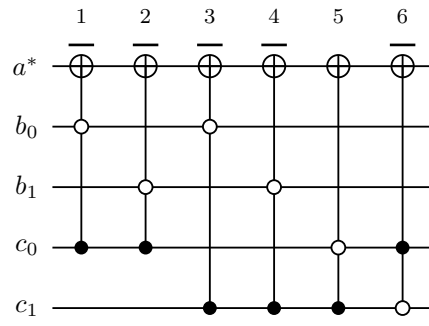
**3 — linear tail**



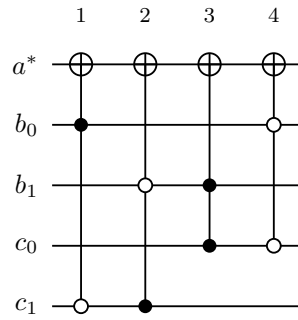
**4 — legacy GADGET**



**5 — quad + pair**



**6 — 4-cube ESOP**



Reading these against the algebra: in 0–2 a CNOT pair collapses a value onto one carrier (the  $\bullet$  on  $b_1/c_1$  writing  $b_0/c_0$ ) and the mirror pair at the end restores it; variant 4 uses  $c_1$  as scratch, written and restored by the identical gates in columns 1 and 4. The masking-safe realizations (3 and 6) are exactly those whose every gate reads at most one carrier of each logical value.